

CE 310

Empirical Percentiles Across CE Practice

Week 6 · Class 1 · Concrete, Traffic, and Cost — No Model Required

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Opener

Who Picks the Percentile?

The 85th-Percentile Rule — A Live Debate, Not a Fixed Fact

For decades, U.S. traffic engineers have set speed limits using one rule: find the speed **85% of free-flowing drivers already travel at or below**, and post that as the limit. The theory: most drivers naturally choose a safe speed, so the 85th percentile reflects sound collective judgment.

In the last decade, that rule has come under real pressure. USDOT's own guidance and organizations like NACTO now argue that in urban, pedestrian-heavy corridors, the 85th-percentile rule can *ratchet speeds upward over time* — every driver who speeds a little pushes next year's "safe" limit a little higher — and that it ignores who's actually at risk when a car and a pedestrian collide.

💡 Neither side of this debate is arguing about how to *compute* a percentile. They're arguing about **which percentile is the right one to build a policy on** — and that's an engineering judgment call, not a statistics problem. This week gives you the tool to compute the number correctly; the judgment is yours.

The Investigation Dataset — Four Engineering Processes

Today's lab spans four independent engineering processes — `Week5_CE_Measurements.csv` plus one new file:

Domain	Variable	Real industry standard
Structural — concrete strength	<code>fc_psi</code>	ACI 318 (10th-pct compliance)
Acoustics — site noise level	<code>NoiseLevel_dB</code>	L10/L90 descriptors, HUD guidance
Transportation — spot speed	<code>Speed_mph</code>	The 85th-percentile speed rule
Cost engineering — unit cost	<code>CostPerCY</code>	AACE contingency (high pct)

Last week: a Normal/lognormal model for concrete, speed, and cost. **This week:** rank and count the *actual* records for all four — no model. (Acoustics is new — purely empirical, no Week 5 comparison.)

Today's Investigation — Three Acts, Four Domains

Today we investigate the data in sequence — each act opens with a real question that demands a tool, rather than listing tools and then applying them.

Act 1 — Concrete: Does Supplier Beta comply with ACI 318? The answer requires ranking real records — no model assumed.

Act 2 — Speed, Noise, Cost: The same two Excel functions answer three more industry-standard questions across completely different domains.

Act 3 — Model check: Does the Normal distribution from Week 5 agree with the raw counts? And does the answer change when you split data into subgroups?

Dataset: Mostly the same as Week 5, plus `Week6_Noise_Measurements.csv` (500 new noise readings).

Act 1

Does This Concrete Comply? — Ranking the Records

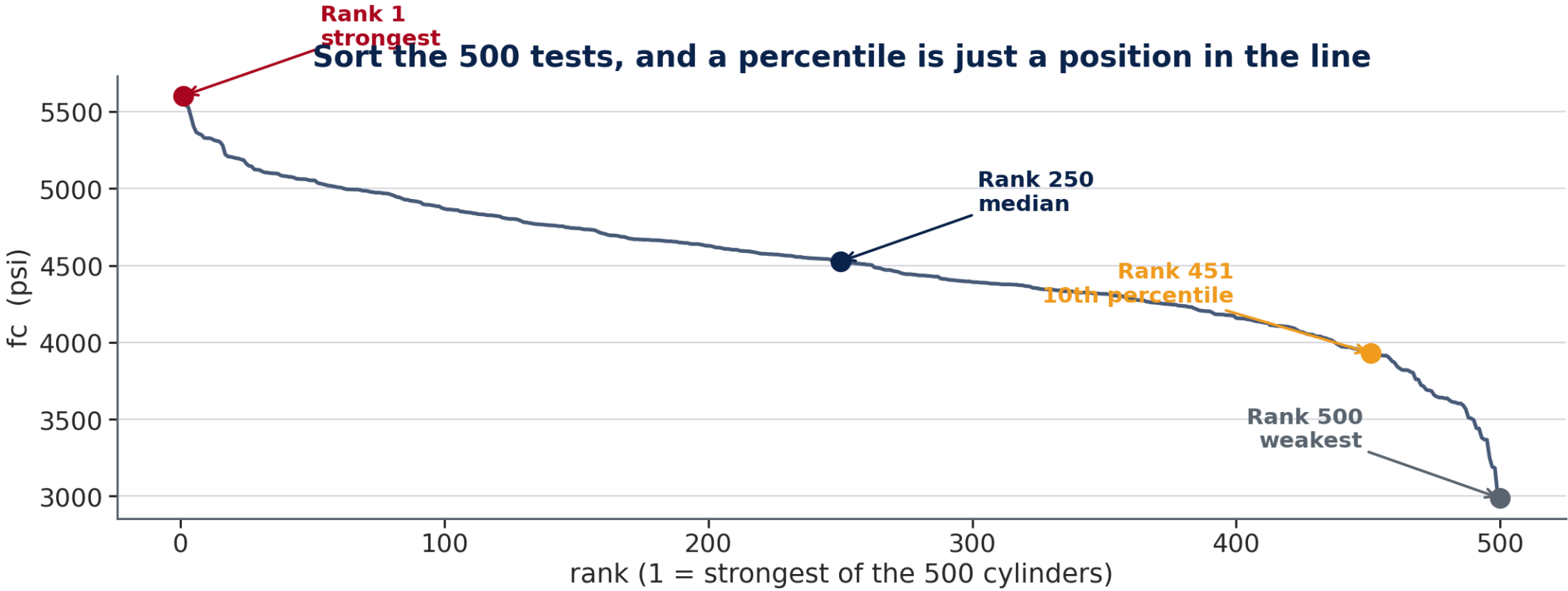
What Does "The 10th Percentile" Actually Mean?

Imagine sorting all 500 concrete cylinder tests from **strongest to weakest**.

- **Rank 1** — the single strongest cylinder tested
- **Rank 50** — the 50th-strongest ($\approx 10\%$ from the bottom, since $50/500 = 10\%$)
- **Rank 250** — the middle value (the median)
- **Rank 500** — the single weakest cylinder tested

A **percentile** is just a rank expressed as a fraction of the total. ACI 318 cares about the **10th percentile from the bottom** — the strength value that only 10% of tests fall below.

A Percentile Is a Position in the Line



LARGE and SMALL — Reading a Rank Directly

=LARGE(range, k) returns the **kth largest** value. =SMALL(range, k) returns the **kth smallest**.

Neither assumes anything about the shape of the data — they just count.

How to choose k for a percentile (n = 500):

$$k_{\text{from bottom}} = \text{ROUND}(n \times p, 0) \quad k_{\text{from top}} = \text{ROUND}(n \times (1 - p), 0)$$

Target	Direction	k	Formula	Result
10th percentile, all concrete	SMALL , k = ROUND(500 × 0.10)	50	50th smallest of the 500	3,930 psi
90th percentile, all concrete	LARGE , k = ROUND(500 × 0.10)	50	50th largest of the 500	5,053 psi

💡 Compare to the interpolated 10th percentile, **3,937 psi** — nearly identical to `SMALL`, because 50 is far enough from the edge of the data to be reliable. The two tools agree; the next slide shows when they don't.

What If the Value Falls Between Two Data Points?

`LARGE` and `SMALL` only ever return a value that actually occurred. But the exact rank you want doesn't always land neatly on one observation — especially when you condition on a subgroup with fewer records.

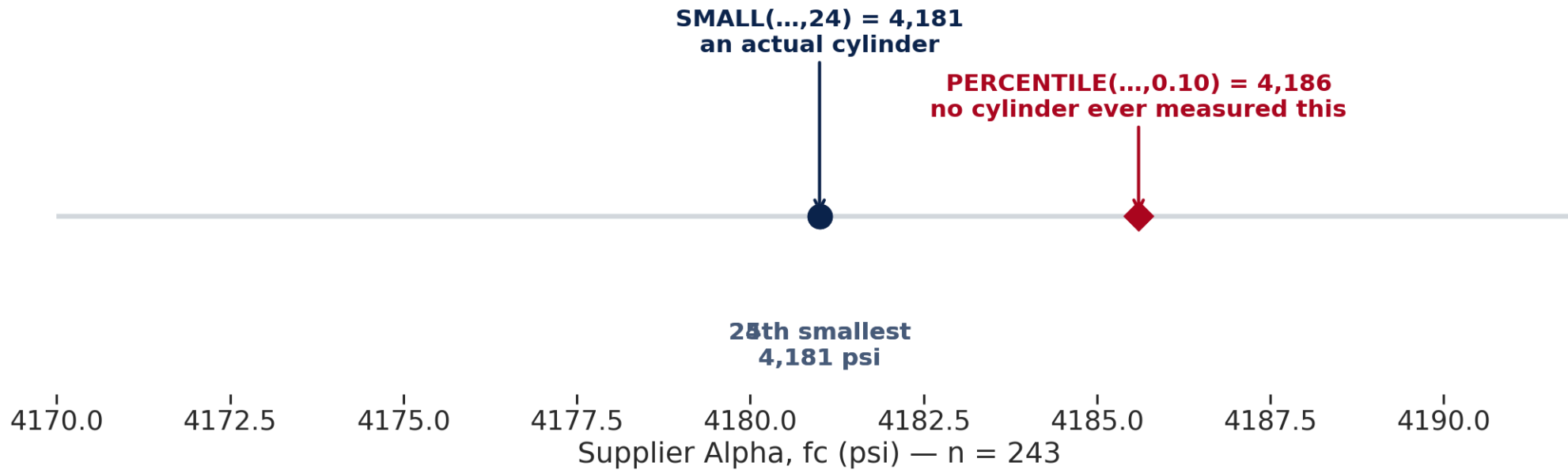
The tool: `=PERCENTILE(array, p)` interpolates linearly between the two nearest sorted values.

Supplier	n	10th pct — SMALL (exact rank)	10th pct — PERCENTILE (interpolated)
Alpha	243	24th smallest = 4,181 psi	interpolated = 4,186 psi
Beta	257	26th smallest = 3,688 psi	interpolated = 3,690 psi

⚠️ ACI 318's own compliance test is written in terms of a fixed **fraction of tests**, so exact-rank ``SMALL`/`LARGE`` is the standard's own method. Use ``PERCENTILE`` when you want a smooth, interpolated estimate instead — both are legitimate, they just answer slightly different questions.

Exact Rank vs. Interpolated — Drawn

Two answers to the same question, both defensible



ACI 318 — Does This Supplier's Concrete Actually Comply?

ACI 318 requires that no more than 1-in-10 concrete cylinder tests fall below the specified strength f'_c — a **10th-percentile compliance test**, written directly into the building code.

Supplier	Empirical 10th pct (SMALL)	Week 5 Normal-model estimate	Gap	Complies with $f'_c = 4,000$ psi?
Alpha (n=243)	4,181 psi	4,191 psi	−10 psi	✓ Yes — comfortably above spec
Beta (n=257)	3,688 psi	3,747 psi	−59 psi	✗ No — below spec

⚠ Beta's empirical 10th percentile is **59 psi lower** than the Normal model predicted — nearly 6× the gap Alpha showed. Beta's data is more variable ($\sigma = 484$ psi vs. Alpha's 344 psi), and the more variable a supplier's output, the less the Normal model's tail estimate can be trusted without checking it empirically.

How Often Does a Threshold Actually Get Crossed?

Exceedance probability: the fraction of records that exceed (or fall below) a threshold — pure counting, no model.

$$P(X < x) = \frac{\text{COUNTIF}(\text{range}, "<" \& x)}{n} \times 100\%$$

The pattern is always the same:

Direction	Excel	Read as
Below threshold	count below x , $\div n$	"% of records below x "
Above threshold	count above x , $\div n$	"% of records above x "

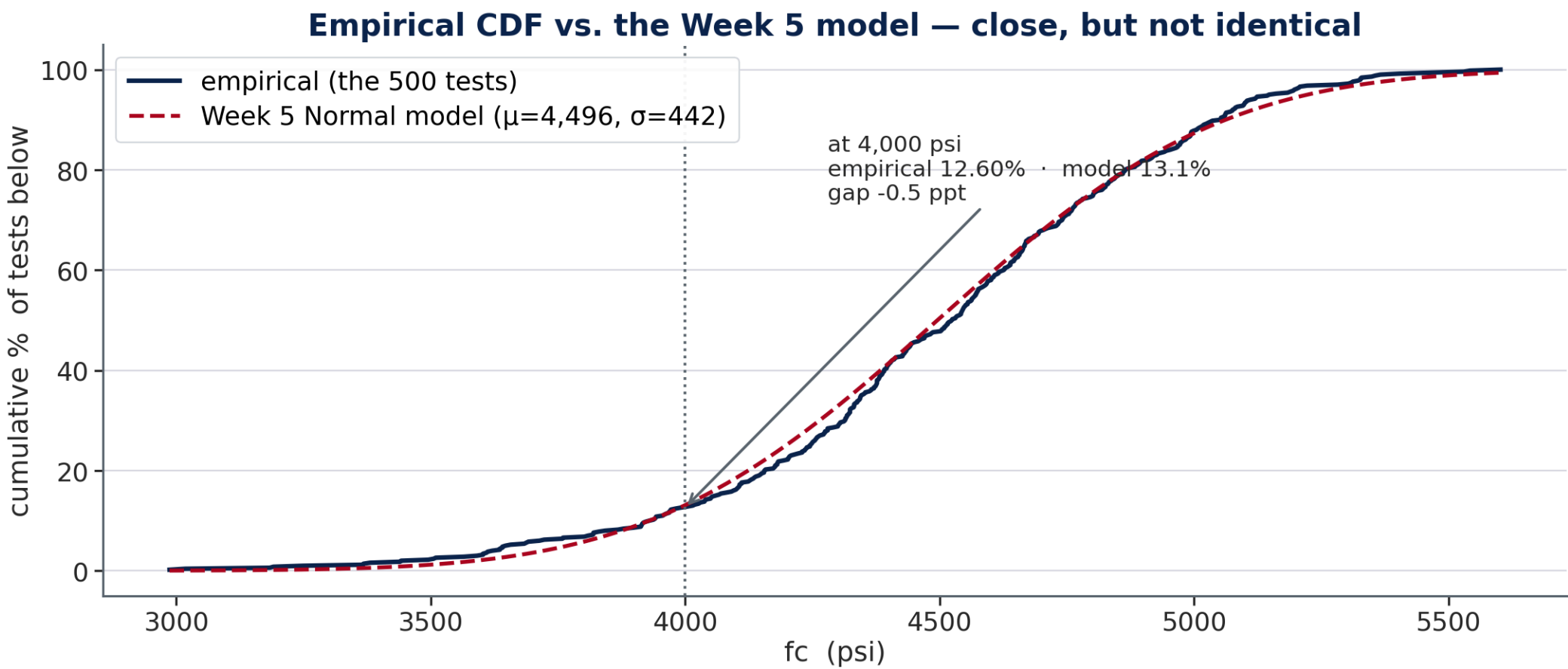
💡 Like `LARGE` and `SMALL`, `COUNTIF` requires no distributional assumption — it counts records that satisfy a condition. The denominator n is always the number of records in the range being counted.

Exceedance — Empirical vs. Week 5 Model

Question	Empirical	Week 5 Normal/lognormal model	Gap
P(concrete < 4,000 psi)	12.60%	13.1%	−0.5 ppt
P(speed > 45 mph)	88.20%	88.5%	−0.3 ppt
P(cost > \$250/CY)	16.80%	15.8%	+1.0 ppt

💡 All three gaps are small — the Normal/lognormal models from Week 5 were reasonable approximations for this dataset. That won't always be true; Act 3 shows exactly where and why the gaps grow.

The Two Curves, Side by Side



What Is a Duration Curve?

A **duration curve** ranks every observation from highest to lowest and plots them in order — a direct visual of the entire empirical distribution, no binning, no model.

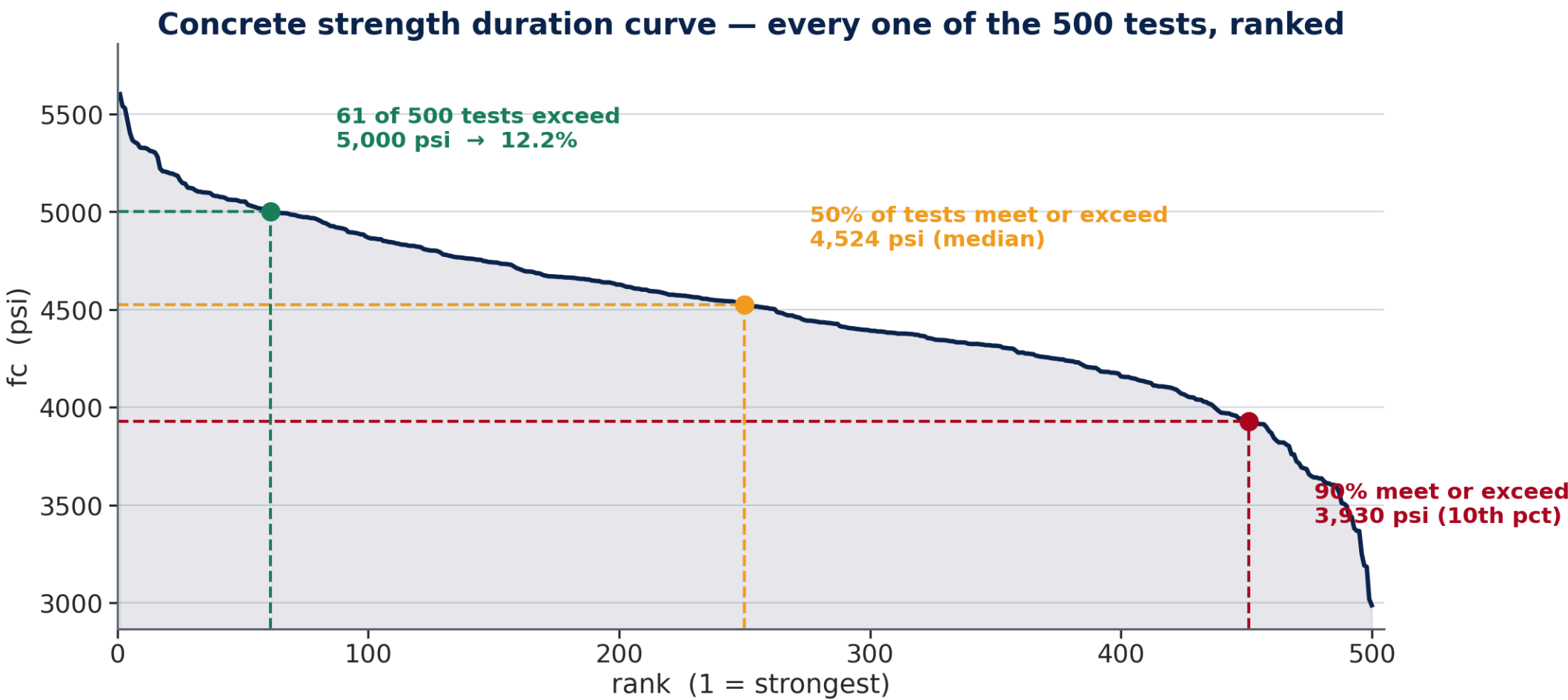
How to read it:

- X-axis: rank (1 = strongest/fastest/costliest, n = weakest/slowest/cheapest)
- Y-axis: the measured value at that rank
- **Any point (x, y)** means: value y is exceeded (or not reached) by x records

Same idea, different name by field:

- Structural/QC engineers: "strength duration curve"
- Traffic engineers: "speed distribution curve"
- Hydrologists: "flow duration curve"

The Duration Curve Itself



Reading the Curve for a Design Decision

Concrete strength, ranked strongest → weakest ($n = 500$):

Design question	Reading the curve	Answer
What strength do 90% of tests meet or exceed?	At rank 450 (90% from top)	≈ 3,930 psi (10th percentile)
What strength do 50% of tests meet or exceed?	At rank 250	≈ 4,525 psi (median)
What fraction of tests exceed 5,000 psi?	At value = 5,000, read the rank	rank 61 → 12.2%

💡 The duration curve is the empirical version of a CDF's tail — but plotted with **rank** on the x-axis instead of cumulative probability, and it works identically whether the underlying variable is concrete strength, vehicle speed, or unit cost.

Act 2

Same Toolkit, Three More Domains

*You've answered the concrete compliance question. The same **LARGE** and **SMALL** functions apply directly to traffic speed, site acoustics, and construction cost — different domains, identical mechanics.*

The 85th-Percentile Speed Rule — Applied to Real Data

Recall the Opener: the 85th-percentile rule sets the speed limit at the value 85% of drivers travel at or below.

Scope	Empirical 85th pct (LARGE)	Week 5 Normal-model estimate	Gap
All records (n=500)	62.1 mph	62.5 mph	−0.4 mph
AM Peak only (n=278)	60.9 mph	60.7 mph	+0.2 mph
PM Peak only (n=222)	63.0 mph	63.6 mph	−0.6 mph

And the exceedance side of the same question — if a 45 mph limit is posted, how many drivers are already over it?

Scope	Empirical P(Speed > 45)	Week 5 Normal-model estimate	Gap
AM Peak	80.94%	81.0%	−0.1 ppt
PM Peak	97.30%	99.2%	−1.9 ppt

How Loud Is "Loud" in Site Acoustics?

Building acoustics uses two named, percentile-based descriptors: **L10** (level exceeded 10% of the time — "peak"/intrusive) and **L90** (exceeded 90% of the time — "background"). Both are just `LARGE / SMALL`.

New file: `Week6_Noise_Measurements.csv` — 500 site readings, `Location` = "Roadside"/"Courtyard", `NoiseLevel_dB`.

Metric	Formula	Value
L10 (peak), all sites	50th largest of 500 readings	67.7 dB
L90 (background), all sites	50th smallest of 500 readings	44.7 dB
HUD check: P(noise > 65 dB)	count above 65, ÷ 500	18.00%

⚠ HUD site-acceptability guidance treats ≤65 dB as "Acceptable." **18% of readings exceed it** — computed with the same two functions you just used for concrete.

AACE Contingency — Budgeting for the Cost You'll Probably Actually Pay

Cost engineers rarely budget to the **mean** — a project where all 500 line items come in "average" is a fantasy. AACE guidance budgets contingency to a **high percentile** instead (commonly P80–P90, by risk tolerance).

Metric	Empirical (LARGE / PERCENTILE)	Week 5 lognormal-model estimate	Gap
90th percentile cost	\$286.50–\$287.56/CY	\$289.61/CY	≈ −\$2–3/CY
P(cost > \$250/CY)	16.80%	15.8%	+1.0 ppt

💡 Budgeting to the mean (\$194.75/CY) instead of the empirical 90th percentile (\$286.50/CY) means a contingency reserve that fails to cover cost roughly **1 in 6 times** — the same "flaw of averages" logic from Week 1, now applied to a cost estimate instead of a temperature.

Act 3

Does the Week 5 Model Hold Up?

A Tale of Three Domains

Both methods describe the **same 500 records**. Where they agree, last week's model was a useful simplification; where they diverge, trust the empirical result. *(Acoustics has no Week 5 parametric model to compare against — it's a pure empirical exercise from the start, which is its own kind of lesson: not every real-world variable gets a convenient textbook distribution fit first.)*

Metric	Empirical	Week 5 model	Gap	Implication
Concrete 10th pct, Alpha	4,181 psi	4,191 psi	−10 psi	Low-variance supplier
Concrete 10th pct, Beta	3,688 psi	3,747 psi	−59 psi	High-variance supplier
Speed P(>45), PM Peak	97.30%	99.2%	−1.9 ppt	Largest gap found
Cost P(>\$250)	16.80%	15.8%	+1.0 ppt	Model understates tail risk

💡 Agreement is closest for **low-variance** subsets (Alpha concrete) and weakest for **higher-variance, smaller subgroups** (Beta concrete, PM Peak speed).

The IQR Check — A Model-Free Diagnostic

The **interquartile range** (IQR = Q3 – Q1) is robust to outliers and describes only the middle 50% of the data — a check on the *shape* of a distribution that needs no assumptions.

For a perfect Normal distribution: $IQR = 1.35 \times \sigma$

Variable	Q1	Q3	Empirical IQR	1.35σ (Week 5 model)	Read
Concrete (all)	4,246.50	4,802.00	555.50 psi	$1.35 \times 442 = 596.7$	IQR narrower — center more concentrated than Normal predicts
Speed (all)	49.00	60.10	11.10 mph	$1.35 \times 7.81 = 10.5$	Close agreement — center is genuinely Normal-like

Predict — Is the Middle Wider or Narrower?

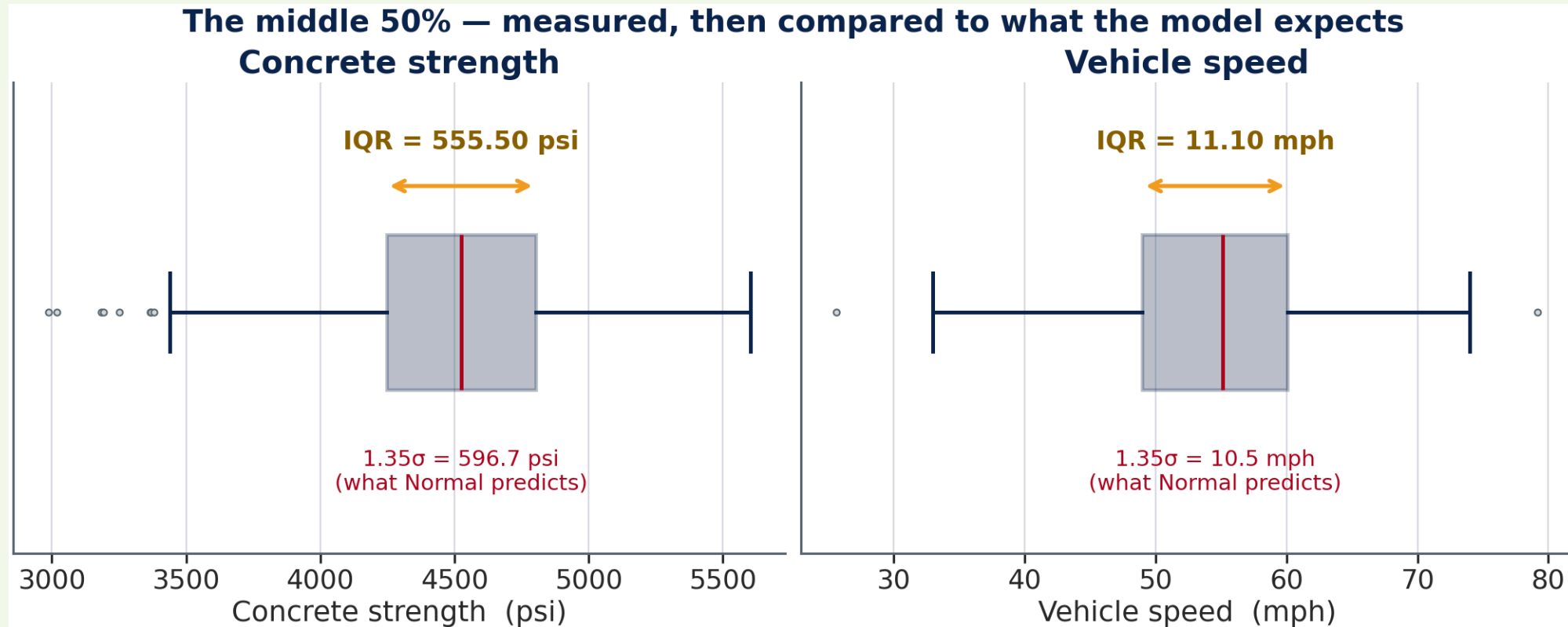
45 seconds

For a perfect Normal distribution, the interquartile range is **1.35σ** .

Week 5's Normal model for concrete strength had $\sigma \approx 442$ psi, so a Normal distribution would put the middle 50% of tests inside a band of about **597 psi**.

Is the real IQR of the 500 tests wider or narrower than that? And what would each answer tell you about the tails?

The Middle 50%, Drawn



Quick Check — Comparing the Two Approaches

Think, then check

You computed for concrete strength:

- Alpha empirical 10th pct: 4,181 psi · Alpha model: 4,191 psi
- Beta empirical 10th pct: 3,688 psi · Beta model: 3,747 psi

Question A: Which supplier's data would you trust the empirical result over the model result *more strongly*, and why?

Question B: A project spec requires $f'_c = 3,800$ psi. Using only the empirical 10th percentiles, does Supplier Beta comply?

✓ Quick Check — Answer

Question A: Supplier Beta. Its empirical-vs-model gap (59 psi) is roughly **6× larger** than Alpha's (10 psi) — and Beta's higher variance ($\sigma = 484$ psi vs. Alpha's 344 psi) is exactly the condition under which a symmetric Normal model is least trustworthy at the tail.

Question B: No. Beta's empirical 10th percentile (3,688 psi) falls below a 3,800 psi spec. Beta fails compliance — and the model-based estimate (3,747 psi) would have failed too, just by a smaller margin. Either way, Beta should not be accepted without further testing.

💡 When the empirical and model-based answers agree on the pass/fail decision (as here), you can be confident either way. The dangerous case is when they disagree on which side of a threshold you're on — always check the empirical number before signing off.

Why Condition? – The Subgroup Effect

The unconditional 85th-percentile speed (62.1 mph, all 500 records) mixes together free-flowing AM commute traffic and denser PM commute traffic. Conditioning on time-of-day reveals the tail behavior for each operating context separately.

Subgroup	85th pct speed	vs. all-records	Interpretation
All records	62.1 mph	—	Mixes both peak periods
AM Peak (n=278)	60.9 mph	–1.2 mph	Lighter traffic, slightly lower design speed
PM Peak (n=222)	63.0 mph	+0.9 mph	Denser but faster-moving corridor traffic

💡 A speed limit set from the unconditional 85th percentile (62.1 mph) is **too low for PM Peak** (where 85% of drivers already exceed 63 mph) and **too high for AM Peak** — pooling the two periods together hides a real operational difference a traffic engineer needs to know about.

Predict — Which Peak Is Faster?

60 seconds · show of hands

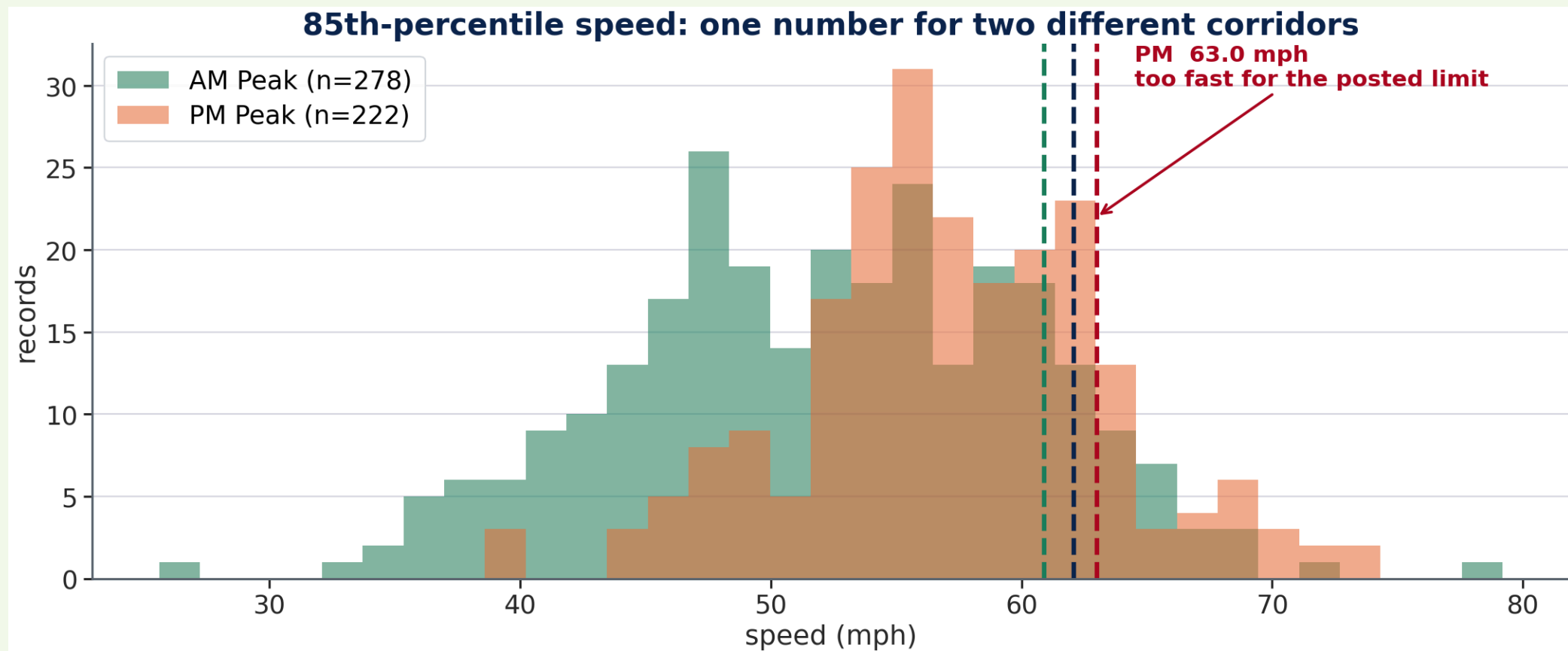
The corridor's **85th-percentile speed across all 500 records is 62.1 mph.**

Split by time of day:

Which peak period has the higher 85th-percentile speed — AM or PM?

Commit to an answer, then estimate the gap in mph.

One Threshold, Two Populations



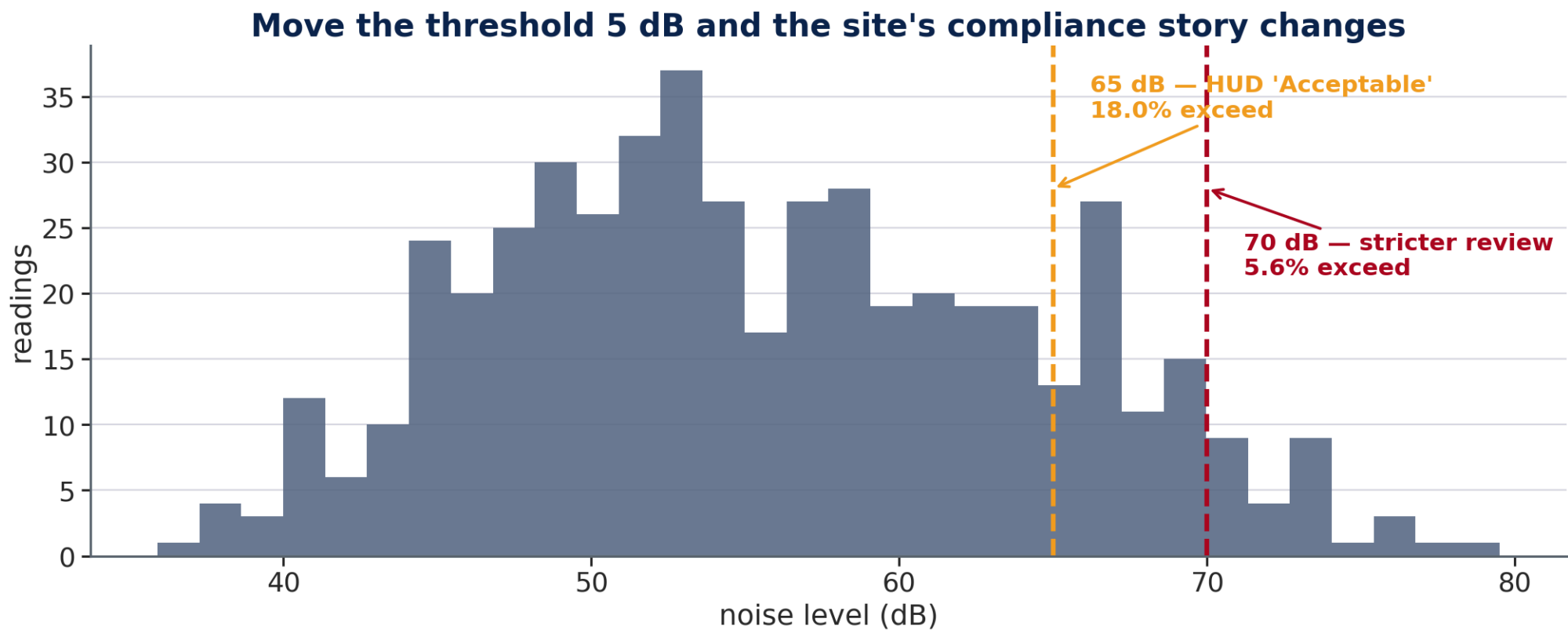
The Same Pattern, One More Domain — Noise by Location

The unconditional L10 (67.7 dB) mixes Roadside and Courtyard-facing sites — two genuinely different acoustic environments.

Subgroup	L10 (peak)	L90 (background)	Interpretation
All sites (n=500)	67.7 dB	44.7 dB	Mixes both site types
Roadside (n=252)	70.7 dB	54.1 dB	Both peak and background are louder
Courtyard (n=248)	54.5 dB	42.4 dB	Much quieter on both measures

💡 Unlike the speed example, Roadside sites are louder on **both** L10 and L90 — the subgroup shifts the entire distribution, not just the tail. A site-acceptability review that pools Roadside and Courtyard readings together would understate risk for Roadside units and overstate it for Courtyard units.

Two Regulatory Lines, One Dataset



Total Cost Exposure Above a Budget Line

Sometimes the engineering question isn't "how often" or "what percentile" — it's **"how much, in total, across every record that exceeds a line?"**

$$\text{Total excess} = \sum_{\text{all records}} \max(\text{Cost} - \text{threshold}, 0)$$

The idea: for every record above the threshold, add up how far above it sits — and ignore the rest.

Threshold	Records above it	Total excess above threshold
\$250/CY	84 of 500	\$5,107.95
\$200/CY	190 of 500	\$11,695.54

💡 This is the same pattern as cooling-degree-hours in a building-energy context, or excess-load summation in a structural load-combination check — `SUMPRODUCT` with a logical condition generalizes to "total exposure above a line" for any engineering threshold problem.

Predict — Does "Mean Plus 15%" Cover It?

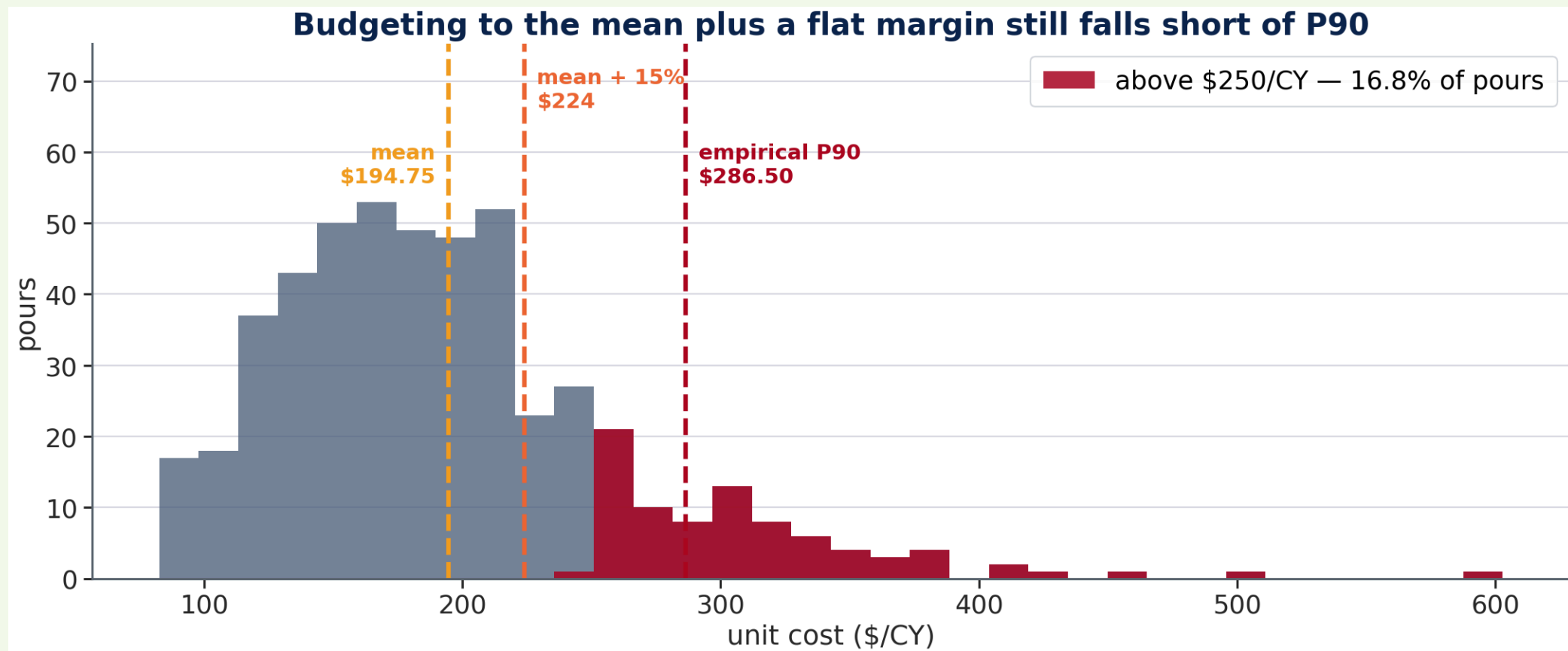
60 seconds · yes or no, then a number

A cost estimator budgets concrete at **the mean, plus a flat 15% contingency.**

For this dataset the mean is **\$194.75/CY**, so the budget line is about **\$224/CY**.

Does that cover the 90th-percentile cost? If not, how far short is it?

Where the Budget Actually Sits



Bridge

Looking Ahead to Regression

AI/ML Bridge — Quantile Regression

Ordinary least squares (OLS), which you meet in Week 9, estimates the **conditional mean**: $E[\text{Cost} \mid \text{predictors}]$. **Quantile regression** estimates a **conditional quantile** instead — e.g., the 90th-percentile cost for a given project.

Why it matters for CE:

- A cost estimator doesn't just want the *average* predicted cost — they want the *90th-percentile* predicted cost, conditioned on project type, size, and location.
- Today's `PERCENTILE(FILTER(...))` is exactly this idea, applied to one categorical predictor at a time.

The connection to today:

Week 6 tool	Quantile regression extension
Unconditional 90th pct (<code>LARGE</code>)	Conditional 90th pct given a predictor
<code>PERCENTILE(FILTER(...))</code>	Conditional by a continuous predictor

Looking Ahead — Week 7: Correlation

You now have three unrelated processes in one dataset — concrete strength, traffic speed, and construction cost, joined only by row number. Next week: how do you tell whether two variables actually move together, versus just happening to sit in the same spreadsheet?

Preview questions:

- Does `fc_psi` predict `Speed_mph`? (It shouldn't — they're unrelated processes.)
- Does `Speed_mph` predict `CostPerCY`? (Also shouldn't.)
- What would it look like in a scatter plot if two variables are genuinely unrelated?

Week 7 tools:

- Pearson r (`CORREL` function)
- Scatter chart with trendline
- Simple linear regression
- R^2 — how much variance does one variable explain in another?

Week 6 Summary

Empirical methods let you check any model against the data it was fit to — no assumptions required

Today's concepts:

- **LARGE / SMALL** — exact-rank values, no distribution assumed
- **PERCENTILE** — interpolated quantiles for values between data points
- **Exceedance probability** — pure counting via **COUNTIF**
- **Duration curves** — the entire distribution, ranked, in one chart

Week 6 Summary — Continued

Today's concepts, continued:

- **Industry standards** — ACI 318 (10th pct), L10/L90 acoustics descriptors, the 85th-percentile speed rule, AACE contingency (high pct)
- **Conditional percentiles** — subgroup effects the pooled number hides
- **SUMPRODUCT** — total exposure above a threshold, across any domain

The core insight: the model is a starting point; the data is the final answer — and the gap between them grows exactly where variance is highest and sample sizes are smallest.

Key Formulas — Rank-Based Tools

Concept	Formula
kth largest/smallest	<code>LARGE(range, k) / SMALL(range, k)</code>
Interpolated percentile	<code>PERCENTILE(range, p)</code>
Exceedance probability	<code>COUNTIF(range, ">x") / n * 100</code>
Conditional percentile	<code>PERCENTILE(FILTER(range, condition), p)</code>

Key Formulas — Aggregation and Diagnostics

Concept	Formula
IQR	<code>PERCENTILE(range, 0.75) - PERCENTILE(range, 0.25)</code>
Total exposure above a threshold	<code>SUMPRODUCT((range-threshold)*(range>threshold))</code>
Normal-model IQR check	$\text{IQR} \approx 1.35 \times \sigma$